Identification of Habitable Exoplanets using Remote Sensing Data

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I. Objective & Methodology

The aim of the study is to identify the most habitable exoplanet, if any, and its conditions for Princess Leia to escape to. The research aims to understand physical conditions of these planets using the data obtained from a remote-sensing satellite. The objective of this research is to analyze the brightness variations of three distant stars, STAC-1, STAC-2 and STAC-3, to detect potential exoplanets using the transit method. The key elements of this study include:

- detect possible exoplanets
- determine their physical and orbital properties
- identify the most promising candidate for further astronomical study
- comparison of the detected planets and identification of the most promising candidate

II. Detection of Exoplanets

A. Transit Method

The transit method is an astronomical technique used to detect exoplanets, which are planets that orbit stars outside our Solar System. When a planet passes in front of its host star, it blocks a small fraction of the star's light, causing a temporary decrease in its observed brightness. By studying these brightness variations over time, we can identify potential exoplanets and estimate their orbital information. When an exoplanet's orbit aligns perfectly with our line of sight from Earth, it passes directly in front of its host star. During this pass, the planet blocks a small fraction of the starlight, causing a temporary dip in the star's observed brightness.

B. Challenges & Limitations

Observational data obtained from satellites was very sparse and limited. Due to these limitations, we weren't able to specifically check for any 'false positive' for any of the three stars or planets. Hence, in the study, we considered all three to be exoplanets due to lack of data and resources to actually check for any false positives or instrumental error. Hence, in the report, we conclude, a transitioning exoplanet in orbits around STAC-1, STAC-2, STAC-3. However, the available observations are insufficient to confirm the presence of an exoplanet or rule out false positives. Further observations are required to verify the periodicity of the planet.

C. Analysis of Observational Data for STAC-1

The revolution of orbiting object shows that the planet shows a constant dip in relative brightness of the star STAC-1 to 0.9970 on every 4th day barring an instance on day-8 when it took 5 days instead of the usual period of 4 days. Also, as stated in Section II.B., due to lack of more data and resources, we can't confirm whether this is due to binary stars or stellar activity or instrumental error. Therefore, we confirm, on the basis of available data and resources, the detection of an exoplanet around STAC-1.

- Calculation of Orbital Period for the Planet:

The orbital period is the time taken by a planet to complete one revolution around its host star. It can be estimated by calculating the time interval between successive brightness dips.

The transits are observed on the following days: Day-3, Day-8, Day-12, Day-16, Day-20.

Taking the duration between each transit, we obtain:

T1= 8-3= 5 days
T2= 12-8= 4 days
T3= 16-12= 4 days
T4= 20-16= 4 days

The orbital period can be calculated as the average of all transit periods.

Orbital Period= $\frac{Sum\ of\ Duration\ of\ Transits}{Total\ Number\ of\ Transit\ Periods}$

Orbital Period= $\frac{5+4+4+4}{4}$ = 4.25 days

Assumption: The observed brightness dips represent successive transits of the same planet. The estimate is approximate because the observations are recorded at discrete time intervals.

- Calculation of Transit Depth:

Transit depth is the fractional decrease in the observed brightness of a star when a planet passes in front of it.

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}}$$

where:
δ = transit depth
F_{normal} = brightness outside transit = 1.000
$F_{transit}$ = brightness during transit = 0.9970
Substituting these values:

$$\boxed{\delta = 0.003}$$

- Conclusion:

In the case of STAC-1, the repeated brightness dips of approximately 0.3%, occurring at roughly regular

intervals, provide evidence for a possible orbiting planet. However, brightness variations alone do not conclusively prove the presence of a planet; additional observations may be required to rule out other possible causes.

D. Analysis of Observational Data for STAC-2

The observations show three decreases in brightness, occurring on days 20, 100 and 180. Each dip has the same recorded brightness value of 0.9990.

The repeated decreases suggest that an object may periodically pass in front of STAC-2, blocking a small fraction of its light. However, the observations alone cannot rule out other causes of brightness variation as stated in Section II.B. Thus, the observations indicate a potential transiting exoplanet, but its presence is not conclusively confirmed.

- Calculation of Orbital Period for the Planet:

The orbital period is the time taken by a planet to complete one revolution around its host star. It can be estimated by calculating the time interval between successive brightness dips.

The transits are observed on the following days: Day-20, Day-100, Day-180.

The interval between successive transits:

T1= 100-20 = 80 days

T2= 180-100= 80 days

The orbital period can be calculated as the average of all transit periods.

Orbital Period= $\frac{Sum\ of\ Duration\ of\ Transits}{Total\ Number\ of\ Transit\ Periods}$

Orbital Period= $\frac{80+80}{2}$ = 80 days

- Calculation of Transit Depth:

Transit depth is the fractional decrease in the observed brightness of a star when a planet passes in front of it.

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}}$$

where:

δ = transit depth

F_{normal} = brightness outside transit = 1.000

$F_{transit}$ = brightness during transit= 0.9990

Substituting these values:

$$\boxed{\delta = 0.001}$$

- Conclusion:

In the case of STAC-2, the repeated brightness dips of approximately 0.1%, occurring at roughly regular intervals, provide evidence for a possible orbiting planet. However, brightness variations alone do not conclusively prove the presence of a planet; additional observations and data may be required to rule out other possible causes.

E. Analysis of Observational Data for STAC-3

The observations show the five decreases in brightness, occurring on days 20, 60, 100, 140 and 170. Each dip does not have the same recorded brightness value.

The first four dips occur at intervals of 40 days, suggesting a possible periodic pattern. However, the fifth dip, on day 170, does not follow this pattern. Therefore, the observations provide some evidence for a potential transiting exoplanet, but the inconsistent final dip introduces uncertainty. The observations alone cannot rule out other causes of brightness variation as stated in Section II.B. Thus, the observations indicate a potential transiting exoplanet, but its presence is not conclusively confirmed.

- Calculation of Orbital Period for the Planet:

The orbital period is the time taken by a planet to complete one revolution around its host star. It can be estimated by calculating the time interval between successive brightness dips.

The transits are observed on the following days: Day-20, Day-60, Day-100, Day-140, Day-170.

The interval between successive transits:

T1= 60-20 = 40 days

T2= 100-60= 40 days

T3= 140-100= 40 days

T4= 170-140= 30 days

The orbital period can be calculated as the average of all transit periods.

Orbital Period= $\frac{Sum\ of\ Duration\ of\ Transits}{Total\ Number\ of\ Transit\ Periods}$

Orbital Period= $\frac{40+40+40+30}{4}$ = 37.5 days

Important limitation: The dip on day 170 occurs only 30 days after the dip on day 140. This does not match the estimated 40-day period. It could be an irregular observation, a separate brightness variation, or evidence that the assumed periodic pattern is incorrect. More observations are required to determine the actual period reliably.

Hence, considering the dip of 170th Day to be due to any other astronomical occurrence, it would be more reliable to take orbital period as 40 days.

- Calculation of Transit Depth:

Transit depth is the fractional decrease in the observed brightness of a star when a planet passes in front of it.

However, since the relative depth in brightness is not equal in all transits, average brightness in transit is calculated.

$F_{transit,avg}$ = $\frac{Sum\ of\ brightness\ in\ every\ transit}{Totsl\ number\ of\ transits}$

$F_{transit,avg}$ = $\frac{0.9990+0.9991+0.9993+0.9992+0.9990}{5}$

$F_{transit,avg}$ = 0.99912

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit,avg}}}{F_{\text{normal}}}$$

where:

δ = transit depth

F_{normal} = brightness outside transit = 1.000

$F_{transit,avg}$ = average brightness during transit = 0.99912

Substituting these values:

$$\boxed{\delta = 0.00088}$$

- Conclusion:

The repeated dips in STAC-3 suggest a possible transit signal, particularly because the first four dips are separated by 40 days. However, the dip on day 170 does not fit the estimated periodic pattern. The small variations in transit brightness and the limited sampling also make it difficult to distinguish a planetary transit from other possible causes. Further observations are necessary to establish whether the dips are genuinely caused by an orbiting planet.

III. Determination Of Planet Size

A. Introduction

The radius of a potential exoplanet can be estimated using the transit depth, which represents the fractional decrease in the observed brightness of its host star when the planet passes in front of it. A greater transit depth generally indicates a larger planet relative to its host star. In this section, the radii of the potential exoplanets orbiting STAC-1, STAC-2, and STAC-3 are estimated using their observed transit depths and the corresponding stellar radii. The calculated planetary radii are expressed in kilometres and in terms of Earth's radius to classify the candidates according to their approximate sizes.

B. Methodology & Equations

The transit depth is related to the radii of the planet and its host star by the following equation:

$$\delta = (\frac{r}{R})\text{^}2 \qquad (1)$$

where,

r = Radius of exoplanet

R= Radius of Star

Constants used in further calculations include:

Earth's radius(R_e) = 6371 km

Solar radius (R_s) = 6,96,000 km

C. Results and Calculations

Assumption: The radius of these exoplanets, in some cases, are very far away from radius of Earth, super-Earth, Neptune or Jupiter, hence, in such scenarios, the planet with closest radius to the exoplanet is chosen.

a. For Star STAC-1

The stellar radius of STAC-1 is 6,96,000 km and its transit dept is 0.003.

Using equation (1), $R_p = R_s \sqrt{\delta}$

$R_p = 696000*\sqrt{0.003}$

$R_p = 696000*0.05477$

R_p = 38,121 km

In terms of Earth's radius:

$R_p = \frac{38121}{6371} * 6371$

$R_p = 5.98\ R_e$

With this radius the planet can be described as Neptune-sized though it is slightly larger than Neptune.

b. For STAC-2

The stellar radius of STAC-2 is 5,56,800 km and its transit dept is 0.001.

Using equation (1), $R_p = R_s \sqrt{\delta}$

$R_p = 556800*\sqrt{0.001}$

$R_p = 556800*0.03162$

R_p = 17,609 km

In terms of Earth's radius:

$R_p = \frac{17609}{6371} * 6371$

$R_p = 2.76\ R_e$

With this radius the planet can be described as Neptune sized though it is significantly smaller than Neptune.

c. For star STAC-3

The stellar radius of STAC-3 is 3,48,000 km and its transit dept is 0.00088.

Using equation (1), $R_p = R_s \sqrt{\delta}$

$R_p = 348000*\sqrt{0.00088}$

$R_p = 348000*0.00088$

R_p = 10,323 km

In terms of Earth's radius:

$R_p = \frac{10,323}{6371} * 6371$

$R_p = 1.62\ R_e$

With this radius the planet can be described as super-Earth.

IV. Determinning Planetary Orbits

A. Methodology & Equations

Kepler's Third Law of Planetary Motion:

$P^2 = a^3 / M$

where,

P = orbital period in years

a = orbital distance in AU

M = mass of the star in solar masses

Rearranging to calculate the orbital distance:

$a^3 = P^2M$

$a = (P^2M)^{1/3}$

Another important conversion factor:

$$1\ AU = 149.6 \text{ million km}$$

The orbital periods estimated in Part A are used below. For STAC-1, the more consistent later transit intervals suggest approximately 4 days; for STAC-2, the estimated period is 80 days; and for STAC-3, the first four dips suggest 40 days, although its periodicity is uncertain.

Also, M∗ denotes the mass of Sun

B. Results and Calculations:

a. For STAC-1:

Orbital Period = 4.25 days = $\frac{4.25}{365.25}$ years = 0.01163 years

Stellar Mass = 1.00 M∗

Calculating orbital distance, $a = (P^2M)^{1/3}$

$a = (0.01163 * 0.01163 * 1)^{1/3} = 0.0513$ AU

The distance between STAC-1 and exoplanet is nearly 5.13% of distance between earth and Sun.

$a = 0.0513 * 149.6 * 10^6$ km $= 7.67 * 10^6$ km

b. For STAC-2:

Orbital Period = 80 days = $\frac{80}{365.25}$ years = 0.21902 years

Stellar Mass = 0.80 M∗

Calculating orbital distance, a = $(P^2M)^{1/3}$

a = $(0.21902 * 0.21902 * 0.80)^{1/3}$ = 0.337 AU

The distance between STAC-2 and exoplanet is nearly 33.7 % of distance between earth and Sun.

a = 0.337 * 149.6 * 10^6 km = 50.42 * 10^6 km

c. For STAC-3:

Orbital Period = 40 days = $\frac{40}{365.25}$ years = 0.10951 years

Stellar Mass = 0.50 M∗

Calculating orbital distance, a = $(P^2M)^{1/3}$

a = $(0.10951 * 0.10951 * 0.50)^{1/3}$ = 0.182 AU

The distance between STAC-3 and exoplanet is nearly 18.2% of distance between earth and Sun.

a = 0.182 * 149.6 * 10^6 km = 27.22 * 10^6 km

V. COMPARING THE HABITABILITY OF EXOPLANETS

A. Aim & Methodology

The aim of this part of study is to identify the most promising candidate for further habitability studies using the estimated orbital distances, stellar luminosities and planetary properties obtained in preceding parts. The analysis also includes a suggestion for an additional observation that could help determine whether the candidate has an atmosphere or might support life.

The methodology to identify the most habitable planet involves analyzing distance of habitable zone from star, stellar luminosity, orbital diistance, planet radius, orbital radius.

The formula used to calculate distance of habitable zone from star is:

$d_{HZ} \approx \sqrt{(L / L\odot)}$ AU

where,

d_{HZ} = approximate distance of the habitable zone from the star

L = luminosity of the star

L⊙ = luminosity of the Sun

B. Identifying the most Habitable Exoplanet:

- Calculating Habitable Zone

1. STAC-1:

L = 1 L⊙

d_{hz} = 1 AU

While orbital distance is 0.0513 AU which is much closer than habitable zone STAC-2:

2. STAC-2:

L = 0.4 L⊙

d_{hz} = 0.632 AU

While orbital distance is 0.0513 AU which is much closer than habitable zone but still better than STAC-1.

3. STAC-3:

L = 0.08 L⊙

d_{hz} = 0.283 AU

While orbital distance is 0.182 AU which is still closer than habitable zone but better than rest of the two stars.

4. On basis of distance of habitable zone from star, STAC-3 seems the best choice followed by STAC-2 and then, STAC-1.

- Comparison of Candidates

Parameter	STAC-1	STAC-2	STAC-3
Stellar Luminosity(L⊙)	1	0.4	0.08
Orbital Distance(AU)	0.0513	0.337	0.182
Habitable Zone(AU)	1	0.632	0.283
Planet Radius(R_e)	5.98	2.76	1.62
Period of revolution(days)	4.25	80	40

- **Most Promising Candidate**

The exoplanet revolving around STAC-3 is the most promising candidate for further habitability studies, based on the available estimates.

The reasons are:

1. Proximity to the habitable zone: Its estimated orbital distance of 0,182 AU is proportionally closer to its approximate habitable-zone distance of 0.283 AU than the other candidates are to theirs.
2. Planetary size: Its estimated radius of 1.62R places it in the Super-Earth size category, making it a useful candidate for further investigation. Having planetary radius comparable to Earth, leads
3. Host-star properties: STAC-3 has a luminosity of 0.08L and a surface temperature of 3800 K. Its lower luminosity means that its habitable zone is closer to the star than the Sun's habitable zone. Its relatively cool host star has a habitable zone closer to the star, making relatively short orbital distances potentially relevant to habitability studies.
4. Presence of atmosphere and water: Its relatively close position to the estimated habitable zone makes it worth investigating whether suitable atmospheric pressure and temperatures could allow liquid water to exist. The current data do not confirm the presence of water.
5. Additional Information to required to check habitability: Transmission spectroscopy could be used to investigate whether STAC-3's candidate planet has an atmosphere. By analyzing starlight passing through the planet's atmosphere during a transit, astronomers may identify gases such as water vapour, carbon dioxide or methane.

The following reasons exoplanets explain why the exoplanets revolving STAC-1 and STAC-2 are not more promising than the exoplanet revolving STAC-3 for habitability:

1. STAC-1

a. Very close to its star: It orbits at approximately 0.049 AU, much closer than Earth is to the Sun, so it may receive excessive heat which is even less than Mercury's distance from Earth.
b. Possibility of high temperatures: Its orbit is far inside the estimated habitable-zone distance of 1.00 AU, making liquid water on its surface and atmosphere unlikely under Earth-like conditions.
c. Large estimated size: Its radius is about 5.98 times Earth's radius, so it may not have a solid, Earth-like surface.

2. STAC-2

a. Inside the estimated habitable zone: It orbits at approximately 0.337 AU, while its estimated habitable-zone distance is 0.632 AU.
b. Potentially too warm: It may receive more stellar energy than a planet in its star's habitable zone, potentially making stable surface liquid water difficult.
c. Relatively large size: Its estimated radius is about 2.76 times Earth's radius, so it may have a thick atmosphere or a composition different from Earth's rocky composition.

These properties make STAC-3 worth investigating further. However, its estimated 40-day orbital period is uncertain because the observed brightness dips are not fully consistent. STAC-3 is a promising candidate for further habitability studies for several reasons. Its estimated radius of 1.62R makes it a potentially interesting Super-Earth-sized planet, while its orbital distance of 0.182 AU is relatively close to the estimated habitable-zone distance of 0.283 AU.

Scientific note: Under the simplified formula in the study, none of the three candidates lies exactly at its estimated habitable-zone distance. STAC-3 is the closest in relative terms, but this does not prove that it is the most habitable. A planet's atmosphere, water availability and the activity of its host star are also important factors.

VI. Conclusion

The findings of the study show exoplanet of STAC-3 to be having highest chances of supporting life and hence, possess highest chances of habitability. Thus, for further investigations and study, investing time, money and resources on STAC-3 and its system would be in the best of interest for all.